Practice workbook · Optimisation part 1: building the formula to maximise or minimise

Using a fixed area, volume or perimeter to remove one variable and show a given formula for the quantity to be optimised.

16 question parts: 1 worked-example retry and 15 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-10

1. Question 8 · part a · 3 marks

2022 JAN · Paper 01

Worked-example retry · Attempt part a only.

8

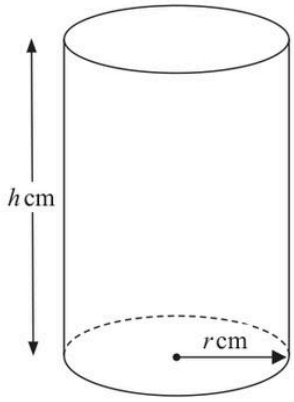


Diagram **NOT** accurately drawn

Figure 1

A solid right circular cylinder has base radius r cm and height h cm as shown in Figure 1.

The cylinder has a volume of $90\pi\,\text{cm}^3$ and a total surface area of $S\,\text{cm}^2$

(a) Show that $S = 2\pi r^2 + \frac{180\pi}{r}$ **(3)**

Given that r can vary,

(b) use calculus to find, to 3 significant figures, the value of r for which S is a minimum, justifying that this value of r gives a minimum value of S **(5)**

(c) Find, to 3 significant figures, the minimum value of S **(2)**

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1. Question 8 · part a · 3 marks · Working and answer

Working and answer

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2. Question 9 · part a · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

9

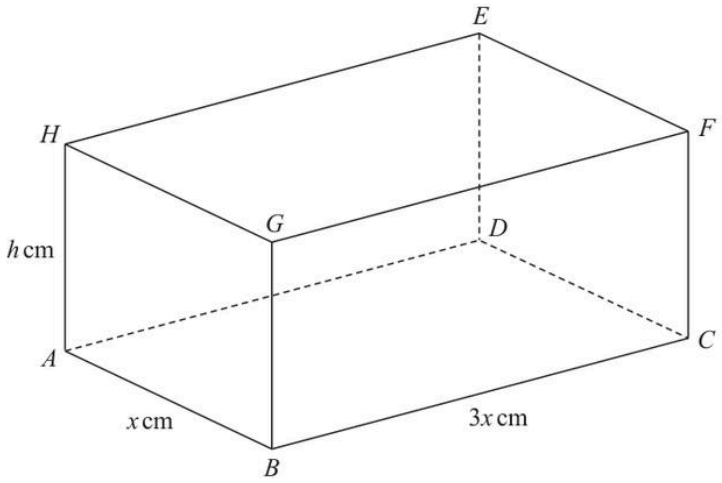


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a solid cuboid $ABCDEFGH$

$AB = x$ cm $\quad BC = 3x$ cm $\quad AH = h$ cm

The volume of the cuboid is 540 cm^3

The total surface area of the cuboid is S cm^2

(a) Show that $S = 6x^2 + \frac{1440}{x}$

(4)

Given that x can vary,

(b) use calculus to find, to 3 significant figures, the value of x for which S is a minimum. Justify that this value of x gives a minimum value of S.

(5)

(c) Find, to 3 significant figures, the minimum value of S.

(1)

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P 6 1 8 8 2 A 0 2 4 3 6

2. Question 9 · part a · 4 marks · Working and answer

Working and answer

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3. Question 8 · part a · 5 marks

2023 JAN · Paper 01

Further practice · Attempt part a only.

8

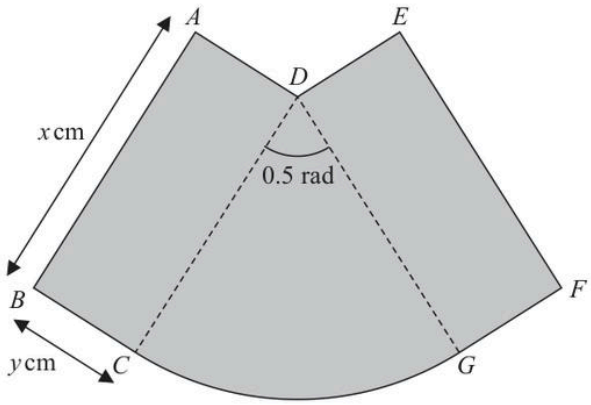


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a badge, shown shaded, made from two identical rectangles, $ABCD$ and $DEFG$, and a sector DCG of a circle with centre D.

Each rectangle measures x cm by y cm.
The radius of the sector is x cm and the angle CDG is 0.5 radians.

The area of the badge is 50 cm^2
The perimeter of the badge is P cm.

(a) Show that

$$P = 2x + \frac{100}{x}$$

(5)

Given that x can vary,

(b) use calculus, to find the exact value of x for which P is a minimum.
Justify that this value of x gives a minimum value for P

(6)

(c) Find the minimum value of P
Give your answer in the form $k\sqrt{2}$, where k is an integer to be found.

(2)

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P 7 1 8 1 7 A 0 2 0 3 2

3. Question 8 · part a · 5 marks · Working and answer

Working and answer

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4. Question 7 · part a · 2 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

7 Two numbers x and y are such that $3x - y = 4$

$S = 5x^3 + y^2$

(a) Show that $S = 5x^3 + 9x^2 - 24x + 16$

(2)

Given that x can vary,

(b) use calculus to find the value of x for which S is a minimum, justifying that this value of x gives a minimum value of S

(5)

(c) Find the minimum value of S

(2)

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4. Question 7 · part a · 2 marks · Working and answer

Working and answer

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5. Question 11 · part a · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

11

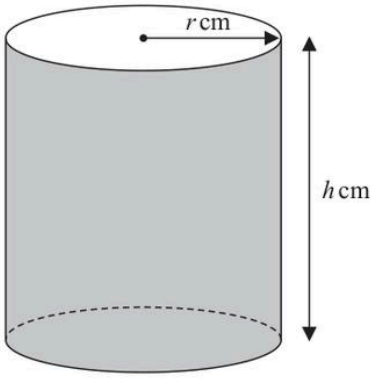


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows an open container in the shape of a cylinder with radius r cm and height h cm.

Given that the total surface area of the container is $625\pi\,\text{cm}^2$

(a) show that

$$h = \frac{625 - r^2}{2r}$$

(3)

The volume of the container is $V\,\text{cm}^3$

Given that r can vary,

(b) use calculus to find the value, to 3 significant figures, of r for which V is a maximum.
Justify that this value of r gives a maximum value of V

(6)

(c) For the value of r found in part (b), find the corresponding value, to 3 significant figures, of h

(1)

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5. Question 11 · part a · 3 marks · Working and answer

Working and answer

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6. Question 7 · part a · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

7

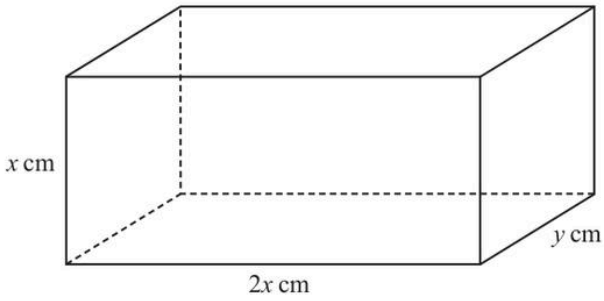


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a solid cuboid with sides of length x cm, $2x$ cm and y cm

The volume of the cuboid is 72 cm^3

The total surface area of the cuboid is A cm^2

(a) Show that $A = \frac{216}{x} + 4x^2$ **(4)**

(b) Use calculus to find the minimum value of A
Justify that your value is a minimum value of A **(7)**

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6. Question 7 · part a · 4 marks · Working and answer

Working and answer

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7. Question 6 · part a · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part a only.

6

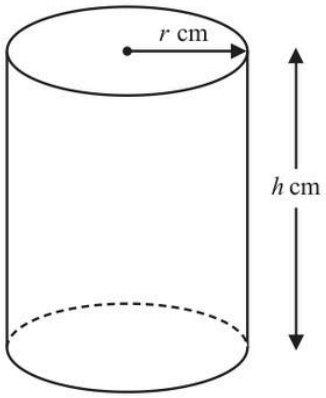


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a solid right circular cylinder with radius r cm and height h cm

The total surface area of the cylinder is 700π cm^2

The volume of the cylinder is V cm^3

(a) Show that $V = \pi r\left(350 - r^2\right)$

(4)

Given that r can vary and using calculus,

(b) find, in cm to 3 significant figures, the value of r for which V is a maximum. Justify that this value of r gives a maximum value of V

(5)

(c) Find, to 3 significant figures, the height h cm for which V is a maximum.

(1)

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P 7 2 8 6 7 A 0 1 2 3 2

7. Question 6 · part a · 4 marks · Working and answer

Working and answer

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8. Question 11 · part a · 6 marks

2020 JAN · Paper 01

Further practice · Attempt part a only.

11

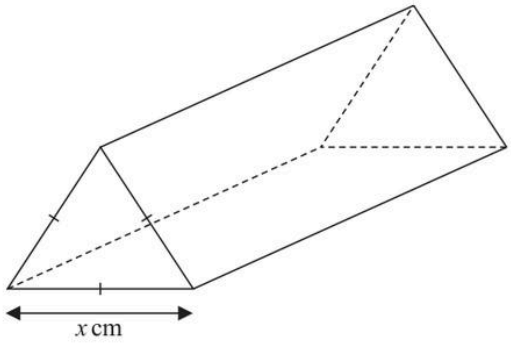


Diagram **NOT** accurately drawn

Figure 4

A company manufactures chocolate bars that are inside packaging that is in the shape of a right triangular prism.

The cross section of the prism is an equilateral triangle with sides of length x cm, as shown in Figure 4.

The volume of the prism is $72\,\text{cm}^3$

The total surface area of the prism is $S\,\text{cm}^2$

(a) Show that

$$S = \frac{\sqrt{3}x^2}{2} + \frac{288\sqrt{3}}{x}$$

(6)

Given that x can vary,

(b) use calculus to find, to 4 significant figures, the value of x for which S is a minimum, justifying that this value gives a minimum value of S.

(5)

(c) Find, to 3 significant figures, the minimum value of S.

(2)

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8. Question 11 · part a · 6 marks · Working and answer

Working and answer

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9. Question 5 · part a · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

5 A solid cuboid has width x cm, length $4x$ cm and height h cm.
The volume of the cuboid is 75 cm^3 and the surface area of the cuboid is S cm^2

(a) Show that $S = 8x^2 + \frac{375}{2x}$

(4)

Given that x can vary, using calculus,

(b) (i) find to 3 significant figures, the value of x for which S is a minimum,

(ii) justify that this value of x gives a minimum value of S

(5)

(c) Find, to 3 significant figures, the minimum value of S

(2)

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9. Question 5 · part a · 4 marks · Working and answer

Working and answer

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10. Question 8 · part a · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

8

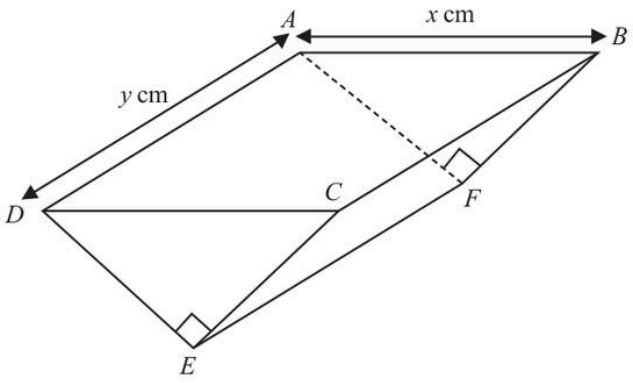


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a solid right triangular prism $ABCDEF$

The cross section of the prism is an isosceles triangle.

- $\angle DEC = \angle AFB = 90°$
- $AB = DC = x$ cm
- $AD = BC = FE = y$ cm
- $AF = BF = DE = CE$

The triangular faces of the prism are vertical and the edges AD, BC and FE are horizontal.
The volume of the prism is 3.6 cm^3
The total external surface area of the prism is S cm^2

(a) Show that S satisfies the equation

$$S = \frac{x^2}{2} + \frac{72\left(\sqrt{2}+1\right)}{5x}$$

(4)

Given that x can vary,

(b) use calculus, to find to 3 significant figures, the value of x for which S is a minimum.
Justify that this value of x gives a minimum value of S

(4)

(c) Hence find, to 2 significant figures, the minimum value of S

(2)

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10. Question 8 · part a · 4 marks · Working and answer

Working and answer

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11. Question 5 · part a · 3 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

5

Diagram **NOT** accurately drawn

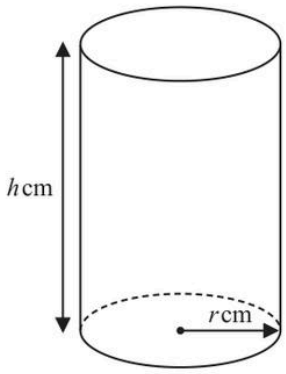


Figure 1

A solid right circular cylinder has base radius r cm and height h cm, as shown in Figure 1

The cylinder has a total surface area of 322π cm^2 and a volume of V cm^3

(a) Show that $V = \pi r\left(161 - r^2\right)$

(3)

Given that r can vary,

(b) use calculus to find, to 3 significant figures, the value of r for which V is a maximum, justifying that this value of r gives a maximum value of V

(5)

(c) Find, to 3 significant figures, the maximum value of V

(2)

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11. Question 5 · part a · 3 marks · Working and answer

Working and answer

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12. Question 11 · part a · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

11

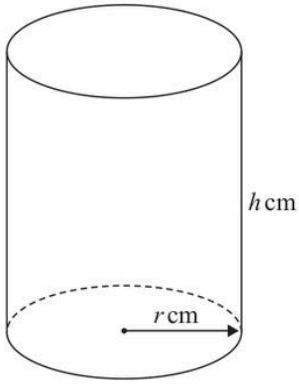


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a solid metal right circular cylinder of radius r cm and height h cm.

The total surface area of the cylinder is $600\,\text{cm}^2$

The volume of the cylinder is $V\,\text{cm}^3$

(a) Show that $V = 300r - \pi r^3$ **(4)**

Given that r can vary,

(b) (i) use calculus to show that the exact value of r for which V is a maximum is

$$r = \sqrt{\frac{100}{\pi}}$$

(ii) justify that this value of r gives a maximum value of V **(5)**

The cylinder is melted down and reformed into a sphere of radius p cm.

(c) Find, to one decimal place, the greatest possible value of p **(3)**

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12. Question 11 · part a · 4 marks · Working and answer

Working and answer

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13. Question 3 · part a · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

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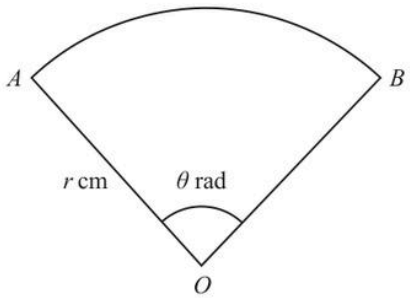


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the sector OAB of a circle with centre O.

The radius of the circle is r cm and the angle AOB is θ radians.

The area of the sector is 675 cm^2

(a) Show that the perimeter of the sector, P cm, is given by

$$P = 2r + \frac{1350}{r}$$

(3)

Given that r can vary,

(b) find, using calculus, the minimum value of P
Give your answer in the form $a\sqrt{b}$ where a is an integer and b is a prime number.

(5)

(c) Justify that the value of P you found in (b) is a minimum.

(2)

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13. Question 3 · part a · 3 marks · Working and answer

Working and answer

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14. Question 7 · part a · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part a only.

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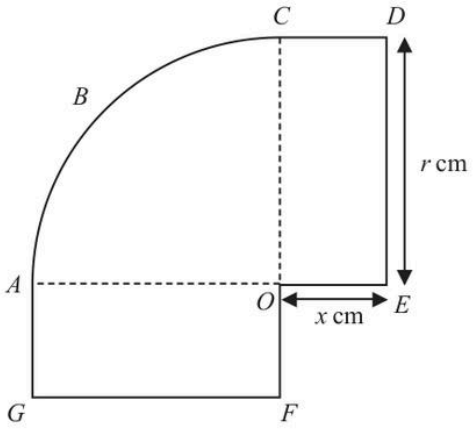


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a shape $ABCDEOFG$

$ABCO$ is a quarter circle with radius r cm
$CDEO$ and $AOFG$ are congruent rectangles of length r cm and width x cm

The total area of the shape is $100\,\text{cm}^2$
The perimeter of the shape is P cm

(a) Show that $P = \frac{200}{r} + 2r$

(4)

(b) Use calculus to find the value of r for which P is a minimum, justifying that this value of r gives a minimum value of P

(5)

(c) Find the minimum value of P

(2)

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P 7 3 5 8 5 A 0 1 8 3 2

14. Question 7 · part a · 4 marks · Working and answer

Working and answer

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15. Question 8 · part a · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

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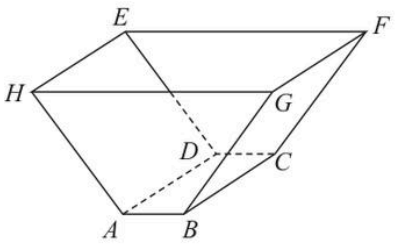


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a waste paper basket in the shape of a right prism with 5 faces and a cross section that is a trapezium. The top, $EFGH$, of the waste paper basket is open.

The base of the prism $ABCD$ is a rectangle with

$$AB = DC = 2x\,\text{cm} \quad \text{and} \quad AD = BC = h\,\text{cm}$$

The cross sections $HGBA$ and $EFCD$ are such that

$$EF = HG = 8x\,\text{cm} \quad \text{and} \quad AH = BG = CF = DE = 5x\,\text{cm}$$

The top, $EFGH$, of the waste paper basket is such that

$$EH = FG = h\,\text{cm}$$

The volume of the waste paper basket is $2250\,\text{cm}^3$

The total surface area of the 5 faces of the waste paper basket is $S\,\text{cm}^2$

(a) Show that $S = 40x^2 + \dfrac{1350}{x}$

(5)

Given that x can vary,

(b) use calculus, to find, to 3 significant figures, the value of x for which S is a minimum.
Justify that this value of x gives a minimum value of S

(5)

(c) Find, to 3 significant figures, the minimum value of S

(2)

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15. Question 8 · part a · 5 marks · Working and answer

Working and answer

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16. Question 7 · part a · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

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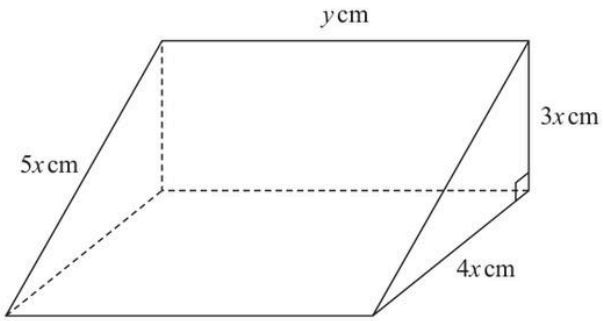


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a block of wood in the shape of a right triangular prism.

The cross section of the prism is a right-angled triangle with sides of length $3x$ cm, $4x$ cm and $5x$ cm.

The length of the prism is y cm.

The total surface area of the five faces of the prism is 144 cm^2

The volume of the prism is V cm^3

(a) Show that

$$V = 72x - 6x^3$$

(5)

Given that x can vary,

(b) use calculus to find the value of x for which V is a maximum, justifying that this value gives a maximum value of V

(4)

(c) Find the maximum value of V

(2)

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16. Question 7 · part a · 5 marks · Working and answer

Working and answer

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